Probability and Probability Distributions

6 (4)

Random Variables

Random Variables

Definition 1:- A random variable is a variable whose value is determined by the outcome of a random experiment.

i.e. a random variable is a numerical variation whose value depends on chance.

Random Variables

Since random variables are outcomes of a random experiment, it is important to understand a random experiment as well. A random experiment is a process which leads to an uncertain outcome.

Random Variables

Tossing (flipping) a coin

For example, when you toss an unbiased coin, the outcome can be a head or a tail. Even if you keep tossing the coin indefinitely, the outcomes are either of the two. Also, you would never know the outcome in advance.

Random Variables

Discrete Random Variables

- ***Discrete random variables*** take on only a countable number of distinct values. Usually, these variables are counts. If a random variable can take only a finite number of distinct values, then it is discrete.

Random Variables

i.e., discrete random variable is a variable that represents numbers found by counting.

For example: number of marbles in a jar, number of students present in a statistics class or number of heads when tossing a coin 5 times etc.

Random Variables

- Number of members in a family, number of defective light bulbs in a box of 10 bulbs, number of houses in the CMC living compound, the number of customers who visit Dashen Bank during any given hour etc. are some examples of discrete random variables.

Random Variables (Contd. . . .)

- ***Continuous random variables*** take up an infinite number of possible values which are usually in a given range. Typically, these are measurements like weight, height, the time needed to finish a task, etc.

Random Variables (Contd. . . .)

For example, the life of an individual in a community is a continuous random variable. Let’s say that the average lifespan of an individual in a community is 70 years.

Therefore, a person can die immediately on birth (where life = 0 years) or after he attains an age of 70 years.

Random Variables (Contd. . . .)

Within this range, he can die at any age. Therefore, the variable ‘Age’ can take any value between 0 and 70 and above.

Some other examples of a continuous random variables are: the weight of a baby, a time taken to complete an exam, the life of a battery.

Probability Distributions

Probability Distributions

Types of probability Distributions:

- Binomial Distribution
- Poisson Distribution
- Normal Distribution

Characteristics of a probability distribution:

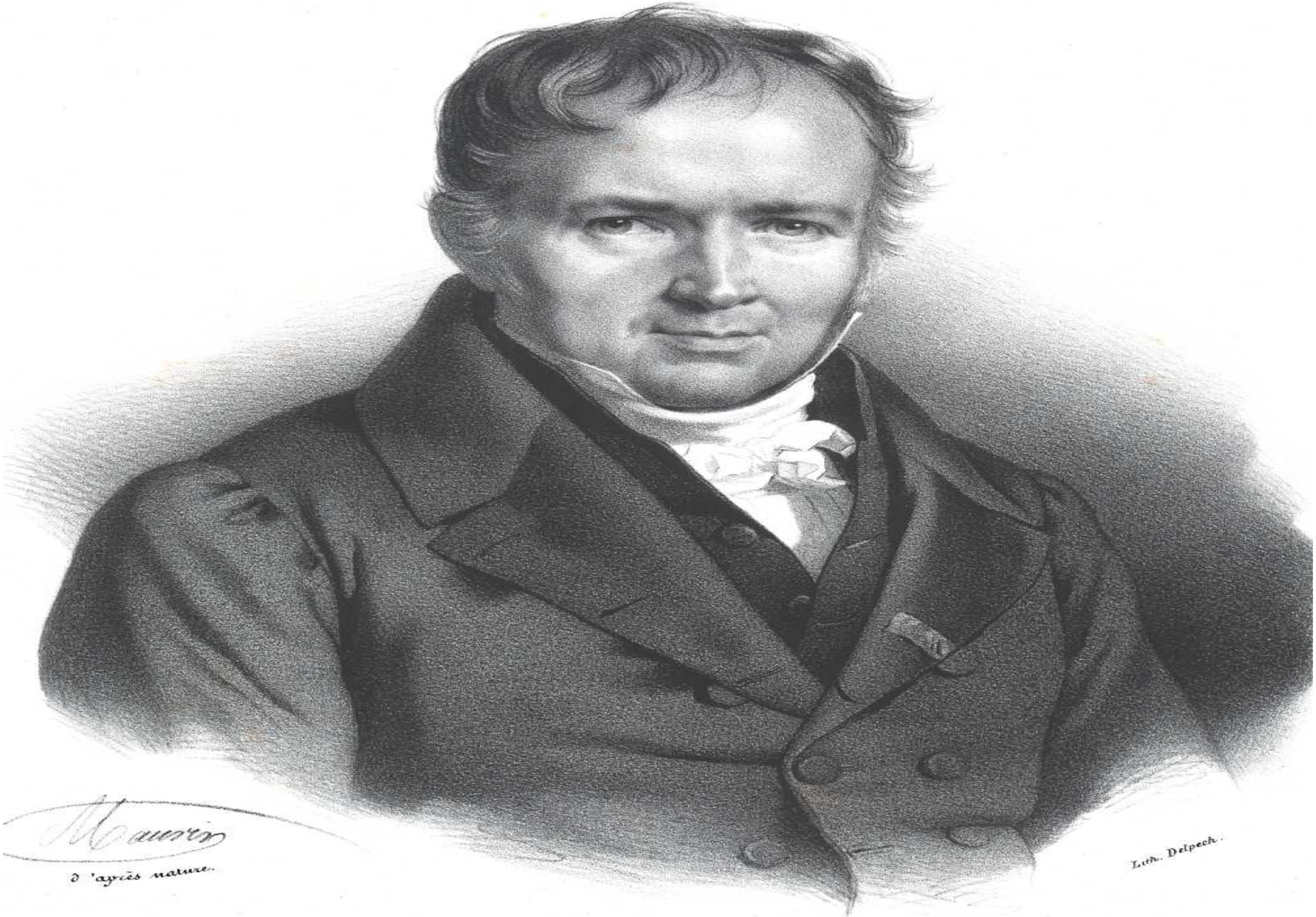
1. $0 \leq p(x) \leq 1$
2. $\sum p(x) = 1$

Binomial Distributions

The prefix 'Bi' means two or twice. A binomial distribution can be understood as the probability of a trail with two and only two outcomes. It is a type of distribution that has two different outcomes namely, 'success' and 'failure'. It is applicable to ***discrete random variables only.***

Poisson Distributions

The Poisson Distribution is a discrete distribution. It is named after Simeon Denis Poisson (21 June 1781 – 25 April 1840) was a French mathematician and physist who worked on statistics, complex analysis, partial differential equations, the calculus of variations and many other mathematical sciences.



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Probability Distributions

The Poisson distribution focuses only on the number of discrete occurrences over some interval. The experiment does not have a given number of trials (n) as binomial experiment does. For example:

- The number of patients arriving at an emergency room within an hour.
- The number of customer support calls received by a call center for 30 minutes.

Probability Distributions

The Poisson distribution and the binomial distribution have some similarities, but also several differences.

The binomial distribution describes a distribution of two possible outcomes designated as successes and failures from a given number of trials.

Probability Distributions

For example, whereas a binomial experiment might be used to determine how many black cars are in a random sample of 50 cars, a Poisson experiment might focus on the number of cars randomly arriving at a car wash during a 20-minute interval.

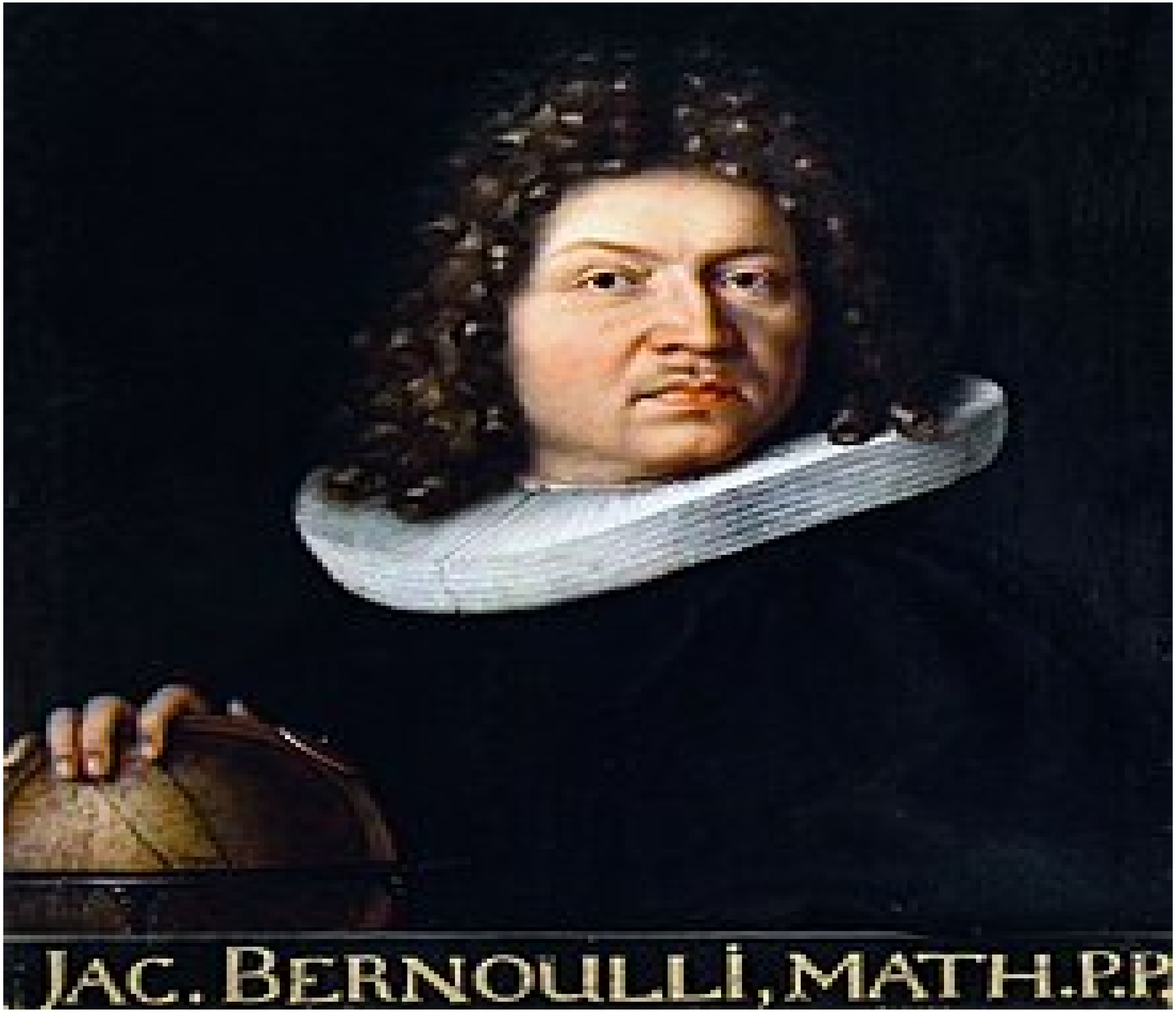
A call center receives on average **6 calls per hour**. What is the probability that **exactly 4 calls** arrive in a **half-hour** period?

Binomial Distributions

Binomial Distributions

- Binomial distribution was discovered by JACOB/JAMES BERNOILLI in 1713.
- This is a discrete probability distribution.
- The discrete probability distribution (applicable to the scenarios where the set of possible outcome is discrete.

Binomial Distributions



Binomial Distributions

- Binomial distributions are used to model situations where there are just two possible outcomes, ***success*** and ***failure***.
- Together with this property, the following conditions also need to be fulfilled,

Binomial Distributions

- There must be a fixed number of trials (called n).
- The probability of success (called r) must be the same for each trial.
- The trials must be independent

Binomial Distributions

Examples where a binomial model could be used.

Trial	Success	Failure
Tossing a coin	Heads	Tails
Throwing a die	Number 6	Not a 6

Binomial Distributions

Example 1:- An unbiased die is thrown ***three*** times and the random variable X denotes the number of sixes obtained. How can we describe the probability distribution X?

Solution 1:- (i) There are a fixed number of trials, that is *three*, each consisting of throwing the die once.

Binomial Distributions

(ii) Each trial has two outcomes since we are looking at the number of sixes obtained. If a six is obtained this is a *success*, and any other outcome is a *failure*.

(iii) The trials are independent. The probability of getting a success, that is a six, is constant for each trial which is $\frac{1}{6}$.

Binomial Distributions

The above experiment satisfies the conditions for modeling by a binomial distribution but what does the probability distribution look like?

We can evaluate the probability by drawing a *tree diagram.*

Binomial Distributions

If X represents the number of success, then

SSS is equivalent to $P(X = 3)$.

$$P(X = 3) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^3$$

and FFF is equivalent to $P(X = 0)$.

$$P(X = 0) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \left(\frac{5}{6}\right)^3$$

Binomial Distributions

What about P(X = 2)? The tree diagram shows that this occurs with SSF, SFS and FSS and these probabilities together are

$$P(SSF) = \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} = \left(\frac{1}{6}\right)^2 \times \frac{5}{6}$$

$$P(SFS) = \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^2 \times \frac{5}{6}$$

Binomial Distributions

$$P(FSS) = \frac{5}{6} \times \frac{1}{6} \times \frac{1}{6} = \left(\frac{1}{6}\right)^2 \times \frac{5}{6}$$

$$\text{So } P(X = 2) = 3 \times \left(\frac{1}{6}\right)^2 \times \frac{5}{6}$$

Now find $P(X = 1)$

Binomial Distributions

So, the probability distribution looks like:

x	0	1	2	3
P(X= x)	$\left(\frac{5}{6}\right)^3$	$3\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2$	$3\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)$	$\left(\frac{1}{6}\right)^3$

Check it is a probability distribution by finding if the probabilities sum to 1.

Binomial Distributions

The dice experiment is an example of a situation that can be modified by a binomial distribution.

Binomial Distributions

Example 2:- The probability of a pregnant woman giving birth to a girl is about 0.49. Draw a tree diagram showing the possible outtcomes if she has two babies (not twins).

Binomial Distributions

From the tree diagram, calculate the probability that:

(i) the babies are both girls,

(ii) the babies are the same sex,

(iii) the second baby is of different sex from the first.

Binomial Distributions

In the dice experiment, the probabilities of the distribution can be calculated because the probabilities are terms in the binomial expansion with $p = \frac{1}{6}$ and $q = \frac{5}{6}$ (where q the probability of failure, that is $q = 1 - p$.)

Binomial Distributions

So generally, the probability that the number of success, X, in n trials is r is given by

$$\mathrm{P}(\mathrm{X}=r)={}^{n}C_{r}\,p^{r}(1-p)^{n-r}$$

Any Questions


Thank
you